

EDUCATION

Carnegie Mellon University <i>Doctor of Philosophy in Electrical and Computer Engineering</i> • Advisor: Olivia Hsu	<i>Expected May 2032</i> Pittsburgh, PA
Cornell University <i>Master of Engineering in Computer Science</i> • Advisor: Adrian Sampson	<i>May 2024</i> Ithaca, NY
Cornell University <i>Bachelor of Science in Computer Science, Electrical and Computer Engineering</i>	<i>December 2023</i> Ithaca, NY

RESEARCH EXPERIENCE

- **Carnegie Mellon University** *May 2026 – Present*
Graduate Student Researcher
Pittsburgh, PA
- **Unaffiliated** *December 2025 – May 2026*
Independent Researcher
Mountain View, CA
 - Developing an interactive synthesis system to schedule a compiler from PyTorch to sparse accelerators [2]
- **Cornell Computer Systems Lab** *January 2022 – May 2024*
Undergraduate Researcher, Capra Research Group
Ithaca, NY
 - Designed compiler from PyTorch to FPGA-executable Verilog with parallel banking optimizations [3] [4]
 - Implemented accelerated pangenomic graph queries using Dahlia, a hardware generation language
- **Purdue Duality Lab** *September 2020 – May 2021*
Undergraduate Researcher
West Lafayette, IN
 - Analyzed domain-specific regular expressions for DFA complexity, portability, and back references
 - Implemented ranking scheme to cluster groups of regexes using metrics such as F-Score and mistake rate

PUBLICATIONS

Refereed Workshop Papers

- [1] **Evan Williams**, Justin Lubin, Rubens Lacouture, and Olivia Hsu. *Interactive Design Space Exploration for Sparse Accelerators*. Young Architect Workshop (YArch) co-located with ISCA. 2026.
- [2] **Evan Williams**, Justin Lubin, Rubens Lacouture, and Olivia Hsu. *Interactive Compiler Scheduling with Strong Guarantees*. Workshop on Languages, Tools, and Techniques for Accelerator Design (LATTE) co-located with ASPLOS. 2026.
- [3] Jiahan Xie, **Evan Williams**, and Adrian Sampson. *From PyTorch to Cayx: An Open-Source Compiler Toolchain for ML Accelerators*. Workshop on Compilers for Machine Learning (C4ML) co-located with CGO. 2026.
- [4] Jiahan Xie, **Evan Williams**, and Adrian Sampson. *From PyTorch to Cayx: An Open-Source Compiler Toolchain for ML Accelerators*. Workshop on Accelerated Machine Learning (AccML) co-located with HiPEAC. 2026.

TEACHING EXPERIENCE

- **18-740: Modern Computer Architecture and Design** *Fall 2026*
Graduate Teaching Intern
Carnegie Mellon University
- **CS 3410: Computer System Organization and Programming** *Summer 2024*
Course Developer
Cornell University
- **CS 3110: Data Structures and Functional programming** *Spring 2022, Spring 2024*
Teaching Assistant
Cornell University
- **ECE 4750/CS 4420: Computer Architecture** *Fall 2023*
Teaching Assistant
Cornell University
- **ECE 5755: Modern Computer Systems and Architecture** *Summer 2023*
Course Developer
Cornell Tech
- **CS 4820: Intro to Analysis of Algorithms** *Fall 2022, Spring 2023*
Teaching Assistant
Cornell University
- **CS 159: C Programming** *Spring 2021*
Teaching Assistant
Purdue University

MENTORING

- **Malena Cochran** May 2026 – Present
Carnegie Mellon University B.S. expected 2028
 - Malena is currently investigating techniques for selecting optimal mappings from sparse kernels to hardware backends in the Mosaic compiler via profile-guided optimizations and cost models

HONORS AND AWARDS

- **Computer and Information Science and Engineering Graduate Fellowship (CSGrad4US)** 2025
National Science Foundation (NSF)
- **Cum Laude** 2023
Cornell University
- **Dean's List** 2021, 2022, 2023
Cornell University
- **Outstanding Undergraduate Course Staff** 2022, 2023, 2024
Cornell University College of Computing and Information Science (CIS)
- **Victor H. and Helen T. Green ECE Scholarship** 2021
Purdue University School of Electrical and Computer Engineering (ECE)

WORK EXPERIENCE

- **Amazon Web Services** August 2024 – April 2026
Software Development Engineer, Amazon HealthOmics
Mountain View, CA
 - Led effort to support CUDA driver integration and implemented type coercion to enable new customer workloads
 - Parallelized integration test execution to reduce time for production deployments in CI/CD pipeline by 50%
- **Amazon Web Services** May 2023 – August 2023
Software Development Engineer Intern, AWS IoT RoboRunner
Sunnyvale, CA
 - Developed application to perform deep packet inspection on MQTT network traffic using C++ and Docker
 - Integrated with Amazon CloudWatch and AWS DeviceDefender to alert customers of anomalous behavior
- **Deloitte Consulting** June 2022 – August 2022
Solutions Engineer Intern
New York, NY
 - Conducted research and drafted design documentation for mobile application to assist crews in electrical outages
 - Composed Python scripts to automate SQL database queries and connected to live front-end using a REST API
- **Commonwealth Associates** May 2021 – August 2021
Electrical Engineering Intern
Jackson, MI
 - Designed alarm modules and relay circuits for electrical substations to improve power and energy efficiency
 - Created mobile application for internal employees to express feedback and ideas to upper-level management

TALKS

- **Compiler Infrastructures for Hardware Accelerator Design** April 2026
Google
- **Frontend Accelerators for Pangenomic Graph Queries** May 2022
NSF PPoS Panorama Meeting for Computational Genomics

SERVICE

- ASPLOS 2027 Artifact Evaluation Committee
- PACT 2026 Artifact Evaluation Committee